Mathematical Specification of the Certificate Software

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1 Purpose and boundary

The installed package `zipper_algebra` studies presentations motivated by zipper relations in quantum topology and their untwisting quotients. Its initial maintained operations construct explicit presentations, expand recorded proofs, compose two-sided ideal-membership certificates, and check the resulting identities exactly. These operations support research on the relations among sliders and on quotients related to BMW algebras. They do not, by themselves, prove finite generation, a dimension formula, termination of the geometric zipper algorithm, or the topological origin of its relations.

This specification describes implemented algebraic operations. The full zipper algorithm, thesis exposition and diagrams are separate work. Historical experiment scripts remain available with their original evidence, but their mathematical claims require individual verification.

2 Ambient algebras and conventions

Put $P = \mathbb{Q}[q,t]$ and $K = \mathbb{Q}(q,t)$, where q, t are algebraically independent central indeterminates. For $n \geq 2$, use the free associative algebra

$$F_n = P\langle x_1, \ldots, x_{n-1}, y_1, \ldots, y_{n-1}\rangle.$$

A word is an ordered tuple of indices in that displayed generator order, starting at one. The empty tuple represents the unit. Concatenation denotes multiplication, in the order written. In the braid group algebra, x_j maps to σ_j and y_j to σ_j^{-1}.

The presentation explicitly includes the inverse relations $x_j y_j - 1$ and $y_j x_j - 1$, adjacent braid relations and far commutations, together with redundant inverse or mixed-sign versions used by the reduction engine. Let J_n be the ideal generated by the braid and inverse relations. Passing from F_n to F_n/J_n identifies freely different words representing the same braid. Consequently, an identity in the braid group algebra need not be a literal free-algebra identity. The maintained verifier checks the latter. A certificate using braid simplification must supply the corresponding braid/inverse terms or use a separately specified group-algebra checker.

3 The presentation builder

The function `build_relations_ring` lives in `src/zipper_algebra/zipper_in_ms.py`. Its front end is the class `Presentation` in `src/zipper_algebra/presentation.py`. A presentation carries

the strand count, untwisting choice, framing and twist powers, level limits, zero-slider relations and named additional relations. Its symbolic relation content is hashed; readable relation labels alone do not identify an algebra.

3.1 Sliders

For a block of i strands starting at j, define

$$U^+_{i,j} = x_j x_{j+1} \cdots x_{j+i-2}, \qquad U^-_{i,j} = y_j y_{j+1} \cdots y_{j+i-2}.$$

The implemented recursion is

$$\begin{aligned} S_{2,j} &= qy_j + (1-q) - x_j, \\ S_{i,j} &= (q^{i-1}U^-_{i,j} - U^+_{i,j})S_{i-1,j} \quad (3 \le i \le n), \end{aligned}$$

with $1 \le j \le n-i+1$. The unshifted slider X_i means $S_{i,1}$.

3.2 Zipper and canonical untwisting relations

The level-two zipper relator is

$$Z_{2,j} = (qy_{j+1} + 1 - q - x_{j+1} - qy_jy_{j+1} + x_jx_{j+1})S_{2,j}.$$

For $i \ge 3$, the emitted higher zipper relator is

$$Z_{i,j} = \big(q^{i-1}U^-_{i,j+1} - U^+_{i,j+1} - q^{i-1}y_jU^-_{i,j+1} + x_jU^+_{i,j+1}\big)S_{i,j}.$$

Here $i \le n-1$ and $1 \le j \le n-i$. These formulas specify the current builder; their equivalence to any alternate geometric formulation is a separate mathematical obligation.

In the canonical untwisting presentation,

$$T_{2,j} = (x_j - t)S_{2,j}, \qquad T_{i,j} = \big(x_{j+i-2}\cdots x_j - q^{i-1}y_{j+i-2}\cdots y_j\big)S_{i,j} \quad (i \ge 3).$$

In the braid quotient, left multiplication by $U^+_{i,j}$ converts the second equation to $(\Theta_{i,j} - q^{i-1})S_{i,j} = 0$, where $\Theta_{i,j} = U^+_{i,j}(x_{j+i-2}\cdots x_j)$. This equivalence uses invertibility; the two expressions are different polynomials in the free algebra.

3.3 Experimental parameter choices

The builder also implements a level-two power relation $(x_j^a - t)S_{2,j}$, the squared relation $(x_j - t)^2S_{2,j}$, and a historical option $(x_j^2 - t^2)S_{2,j}$. The twist-power parameter affects level three only: for exponent $m > 1$, the emitted relation is $(\Theta^m_{3,j} - q^{2m})S_{3,j}$. At higher levels the canonical relation remains. The software does not yet accept an arbitrary untwisting polynomial as input. No structural theorem about a variant follows simply from choosing a flag.

The public presentation object rejects unknown untwisting choices, noninteger strand counts and powers, level bounds outside $2, \ldots, n$, repeated additional relation labels and targets requiring more strands than are present. Numeric coefficients use either $\mathbb{Q}$ or $\mathbb{Z}/m\mathbb{Z}$. Calling the latter a field requires the additional hypothesis that m is prime; the coefficient backend does not perform a primality proof. Nonunits cannot be inverted.

4 Certificate data and exact conclusion

Let $r_1, \dots, r_s \in F_n$ be the ordered defining relations of a specified presentation. A certificate for $f \in F_n$ is a finite list of terms $(u_\nu, i_\nu, v_\nu, N_\nu, e_\nu)$, together with an ordered family of nonzero polynomials $a_1, \dots, a_h \in \mathbb{Z}[q,t]$. Here u_ν, v_ν are words, i_ν names an actual relation, $N_\nu \in P$, and $e_\nu \in \mathbb{N}^h$. The intended identity is

$$f = \sum_\nu \frac{N_\nu}{D_\nu} u_\nu r_{i_\nu} v_\nu, \qquad D_\nu = \prod_{k=1}^{h} a_k^{e_{\nu k}}, \tag{1}$$

in $K\langle x_1, \dots, y_{n-1}\rangle$. Atom irreducibility is unnecessary for checking this identity; it is relevant to reconstruction efficiency. The atom search bounds do not limit the exponents in a composed certificate.

Define $E_k = \max_\nu e_{\nu k}$, with zero maxima for an empty certificate, and $M = \prod_k a_k^{E_k}$. Since P is an integral domain and each atom is nonzero, $M \neq 0$. The exact checker compares the polynomials

$$Mf \quad \text{and} \quad \sum_\nu N_\nu \left(\prod_k a_k^{E_k - e_{\nu k}} \right) u_\nu r_{i_\nu} v_\nu. \tag{2}$$

Equality proves (1) and hence membership of f in the two-sided ideal generated by the relations over K. It also proves membership in the corresponding localization of P, inverting the used atoms. A specialization of the rational identity is available wherever its displayed denominators are nonzero. The polynomial identity (2) remains meaningful at their zeros, but need not imply the specialized membership statement there.

The certificate boundary rejects zero atoms, nonintegral atom coefficients, negative or noninteger polynomial exponents, denominator vectors of the wrong length, inexact numerical coefficients, out-of-range word indices and unknown relation labels. Both verifiers validate the supplied presentation identity. Objects are revalidated at the checking boundary because their data are mutable. A malformed certificate raises a validation error. A well-formed false identity returns a report with `pass=False`.

Legacy files without a presentation hash require an explicitly supplied presentation. Their historical origin does not replace verification against those actual relations. New certificates should retain the presentation hash and explicit target. Named sliders, additional named targets and direct polynomial targets share the same validated target resolution.

5 Independent polynomial identity testing

The second checker evaluates the cleared identity on a product grid of distinct rational integers. For each parameter, its bound includes both the target degree plus the clearing-factor degree and the degrees of every cofactor term. If the bounds are (d_q, d_t), a grid with $(d_q + 1)(d_t + 1)$ points suffices.

Proposition 1. *If a polynomial in $\mathbb{Q}[q,t]$ of bidegree at most (d_q, d_t) vanishes on such a product grid, it is zero.*

Proof. Fix a grid value of t and apply the univariate root bound in q. Each coefficient in q then vanishes at all $d_t + 1$ values of t, so the same root bound makes each coefficient zero. Apply this to every free-word coefficient. □

The checker evaluates the polynomial product M/D_ν directly, rather than dividing its evaluations. Thus grid points at poles cause no undefined division. Negative numerator exponents are rejected: a Laurent polynomial cannot use this polynomial degree bound without additional clearing. This exact finite-grid argument differs from checking a few selected parameter values. Its arithmetic is separate from the symbolic accumulation, but it shares the presentation builder, input validation and target resolution.

6 Proof expansion and contextual composition

A basis proof is an acyclic directed graph expressing each basis element as a sum of contextual multiples of input relations and other basis elements. Forward references are allowed; cycles and missing dependencies are errors. Substitution respects multiplication order. If

$$g_k = \sum_\nu c_{k\nu} L_{k\nu} r_{i_{k\nu}} R_{k\nu}, \qquad f = \sum_s m_s u_s g_{k_s} v_s,$$

composition produces terms with coefficient $m_s c_{k_s\nu}$ and contexts

$$(u_s L_{k_s\nu},\ i_{k_s\nu},\ R_{k_s\nu} v_s).$$

No commutation or cancellation of free words is implicit. Repeated uses of one basis element remain separate until their full contexts agree and coefficients can be added. The number of multipliers must equal the number of skeleton steps, and an explicit nonzero skeleton remainder precludes membership composition.

For sampled multipliers, each sample reduction must have the same ordered sequence of (u_s, k_s, v_s) as the anchor, where k_s is the leading-word key rather than the engine's changing numerical index. The data retain one multiplier per occurrence. A sample with a different contextual schedule is excluded. A successful fit is a candidate; the resulting identity still needs exact verification.

7 Exact reconstruction of candidates

A sample consists of a distinct parameter point in $\mathbb{Q}^2$ and an exact rational value. Repeated locations count once; conflicting values and floating-point coefficients are rejected. For a fixed nonzero denominator proposal and a numerator bidegree bound, interpolation uses the available fitting rows and requires full column rank. A consistent rank-deficient system does not determine a unique numerator and is not completed by arbitrarily setting free variables to zero. Numerator coefficients belong to $\mathbb{Q}$, rather than necessarily $\mathbb{Z}$.

Univariate rational reconstruction fixes a monic denominator, solves the exact linear system, cancels a polynomial greatest common divisor and normalizes the result. The reduced denominator must be nonzero at every supplied point, and the candidate must match every fitting and held-out value. This prevents a spurious cross-multiplied equation $0 = 0$ from concealing an incorrect sample at a pole.

Held-out locations are reserved before fitting and denominator inference. Their coefficients do not participate in inference of the factor family. A passing held-out check is additional evidence, not a rational-function identity proof. Line restrictions can cause distinct polynomial factors to coincide, so inferred exponents are heuristic proposals. They are validated and have a general-search fallback. For arbitrary declared factors, absence of a cancellable whole factor does not imply

coprimality of the numerator and denominator; stronger assertions require irreducible, pairwise nonassociate factors.

Failed predictions, an unavailable common reduction shape and exhaustion of the chosen degree or factor bounds do not show that more data cannot help. The experimental driver retries within its sampling cap and reports an inconclusive outcome when it lacks an exact certificate. In particular, reducing to a nonzero remainder against an uncertified basis establishes no nonmembership conclusion, even at that specialization. The historical exception name `WrongClass` is retained as a subtype of `NeedMoreSamples`; it is an inconclusive diagnostic. Candidate reconstruction is characteristic zero.

8 Evidence and unsupported conclusions

The historical large free-algebra certificate is retained for compatibility checks. Smaller braid-algebra certificates use a different ambient equality and must be described accordingly.

A finite computation, a small residual, agreement across specializations, or a successful process exit does not certify a completed generic Gröbner basis. A nonzero specialized remainder cannot alone establish generic nonmembership. Dimension and trace claims require their own basis-completion and exact-arithmetic evidence. Exact coefficient reconstruction is a maintained candidate-producing operation. Solver execution, generic basis certification and the experiment drivers retain separate completion requirements and remain experimental.

9 Verification commands and source map

The solver-free regressions and distribution checks cover strict certificate validation, noncommuting contexts, repeated basis occurrences, schedule alignment, poles, target degrees and historical certificate compatibility. See `docs/TESTING.md` for the maintained commands and optional integrations.

The maintained implementation is under `src/zipper_algebra/`:

- Presentation and relation formulas: `presentation.py`, `zipper_in_ms.py`.
- Certificate data: `cofactor/cofactor_io.py`, `cofactor/atoms.py`.
- Symbolic and grid checks: `cofactor/verify.py`.
- Proof graph and reduction: `cofactor/proof.py`, `cofactor/skeleton.py`.
- Symbolic composition: `cofactor/compose.py`.
- Coefficient reconstruction: `cofactor/lift.py`.

The experimental sampling driver is `tools/cofactor/pipeline.py`.